MILITARY SPECIFICATION

DEFIBRILLATOR, MONITOR-RECORDER

(Battery Operated, with Charger)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a battery powered defibrillator/monitor-recorder with battery charger suitable for use on patients experiencing cardiac crisis in military combat zone environments.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center ATTN: Directorate of Medical Materiel DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 6515

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SPECIFICATIONS

Military

MIL-T-28800	General Specification for Test Equipment for Use with Electrical and Electronic Equipment.
MIL-D-42048	Deployable Medical Systems, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Heart Association (AHA)

AHA publication titled, "Recommendations for Standardization of Leads and of Specifications for Instruments in Electrocardiography and Vectorcardiography."

(Application for copies should be addressed to the American Heart Association, 7320 Greenville, Ave. Dallas, Texas 75231)

American National Standards Institute\Association for the Advancement of Medical Instrumentation (ANSI\AAMI)

ANSI\AAMI DF2 - American National Standard for Cardiac Defibrillator Devices, with the exception of paragraph 3.2.1.2, Energy Range.

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ANSI\AAMI ES1 - 1985 American National Standard, Safe Current Limits for Electromedical Apparatus.

ANSI/AAMI EC13-1984 American National Standard for Cardiac Monitors, Heart Rate Meters and Alarms.

AAMI ECGC 5\83 - Standard for Electrocardiograph Connectors

(Application for copies should be addressed to the Association for the Advancement of Medical Instrumentation, 1901 North Fort Myer Drive, Suite 602, Arlington, Virginia 22209)

Underwriters Laboratories, Inc.

UL 544 - Standard for Safety of Electric Medical and Dental Equipment.

(Application for copies should be addressed to the Underwriters Laboratories, Inc. Publication Stock, 333 Pfingsten Road, Northbrook, Illinois 60062)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Application. The device shall be a modular or integral defibrillator/monitor-recorder system, suitable for use on patients experiencing cardiac crisis in military combat zone environments. Shall be highly portable and operable from self-contained batteries. Units which are modular in design and can be separated shall consist of only two separate components: a defibrillator and a monitor-recorder. The device shall be capable of synchronous cardioversion. Shall possess a monitor for examining the electrical activity of the heart and a means to produce a permanent record of the heart activity in the three standard ECG lead format. The system shall include batteries, defibrillator paddles, syncronized cardioversion capability and accessories as specified herein. A battery charger must be included as a separate item as part of the package for each defibrillator/monitor-recorder unit.

3.2 Weight. The total weight of the ECG monitor-recorder and defibrillator including the batteries, paddles, etc., shall not exceed 30 pounds.

3.3 Environmental requirements. Shall be capable of full operation while still providing environmental protection and shall comply with the environmental, reliability and maintainability requirements of MIL-T-28880 for type III, class 3, style D, except as specified below:

Atmospheric Pressure:	400 mmHg - 700 mm Hg
Relative Humidity (R.H.)	95 percent (non-condensing)
Temperature:	
Operational	32°F (0°C) to 114°F (45°C)
Storage	-23°F (-30°C) to 149°F (65°C)

3.4 Special tools. Shall require no special tools to assemble/disassemble nor to place into operation.

3.5 Modular units. Units that are modular in design and capable of the monitor and recorder being separable from the defibrillator shall operate independently via integral batteries. Units that are modular shall securely interconnect without the use of auxillary cables and in such a manner as not to cause excessive stress on the electrical connectors.

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3.6 Cleaning. Shall be easily cleaned with routine hospital cleansing solutions and shall be resistive to moisture and cleansing agents, and adequately protected from moisture.

3.7 Ruggedness. Shall be designed to withstand normal abuse such as shock and impact frequently encountered outside the normal hospital environment. The unit must be capable of being fully operated while still providing environmental protection from moisture, dust, or mud.

3.8 Litter compatability. The defibrillator/monitor-recorder is intended for use during the transport of a littered patient and shall be capable of being placed on a litter with a patient while permitting observation at the ECG monitor by one litter bearer.

3.9 Test load. Shall contain an integral 50 ohm test load which may be incorporated in either the defibrillator module or charger module.

3.10 Low battery warning. Each module shall incorporate a visual and audible low battery warning.

3.11 Performance of the defibrillator section. The defibrillator section shall be capable of the performance requirements as specified herein.

3.11.1 Number of discharges. The defibrillator shall be capable of delivering 30 Emax discharges without charging batteries.

3.11.2 Synchronized discharge. The defibrillator shall be capable of syncronizing the defibrillator pulse to a patient-generated ECG "R" wave.

3.11.3 Switch from synchronized mode. The defibrillator shall automatically switch back to defibrillation mode after each cardioversion attempt.

3.11.4 Dual discharge controls. There shall be dual discharge pushbutton controls, one on each paddle, which must be activated simultaneously to discharge the unit.

3.11.5 Remote charge button. There shall be a remote charge button on one paddle handle.

3.11.6 Paddle storage. The defibrillator shall have out-of-the way storage of the defibrillator paddles.

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3.11.7 ECG pickup through paddles. The defibrillator paddles shall also serve as ECG electrodes.

3.11.8 Pulse shape. The defibrillator pulse shape shall be a clinically proven one such as a Lown, Edmark or Trapezoidal, as defined in AAMI DF2.

3.11.9 Pulse energy. The delivered pulse energy shall be within 4 J or plus or minus 15 percent of the selected energy on the energy level indicator, whichever is greater, when discharged into a 50 ohm resistive load.

3.11.10 Maximum energy. The maximum selectable delivered energy of the defibrillator shall be in the range of 320 to 400 joules (J) inclusive. The defibrillator shall provide a selection of at least six energy outputs within the range of 5 J to maximum energy when delivered into a 50 ohm resistive load.

3.11.11 Energy determination. There shall be a visible means of determining selected energy by a calibrated knob setting, digital display, or analog meter.

3.11.12 Ready indication. There shall be a visual ready indicator and audible tone to indicate when the defibrillator is fully charged and ready for discharge.

3.12 Performance of the monitor-recorder section. The monitor-recorder section shall be capable of the performance requirements as specified herein.

3.12.1 Display. The display shall be non-fade, measuring no less than 36 mm high by 60 mm wide.

3.12.2 Frequency response. The monitor shall display a signal with a measurable low frequency cutoff point (-3db down) at 0.5 Hz, and a high frequency cutoff point (-3db down) anywhere between 35 to 50 Hz, inclusive.

3.12.3 Lead selector switch. The operator shall, by means of a lead selector switch, select leads I, II, and III without switching patient electrode sites.

3.12.4 Sweep speed. The monitor shall have a sweep speed of 25 mm/second.

3.12.5 ECG gain. The ECG gain shall be continuously or discrete step mode adjustable.

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3.12.6 Systole indicator. The monitor shall have an easily adjustable audible systole indicator.

3.12.7 Patient cable connector. The patient cable connector shall be a 3 lead/6 pin type configured in accordance with AAMI ECG 5/83 Standard for ECG connectors.

3.12.8 Patient cable. The patient cable shall be fully shielded and 10 feet (3.03 m) long. Individual lead length shall be approximately 3 feet (0.91 m).

3.12.9 Operating time. The monitor-recorder shall operate continuously for a minimum of 2.25 hours of cardioscope operation or 45 minutes of recorder operation (or any linear combination) without use of an external power source.

3.12.10 Input circuit isolation. The input circuit shall be electrically isolated in accordance with the latest edition of the American National Standard, Safe Current Limits for Electrical Apparatus, ANSI/AAMI ES1-1985.

3.12.11 Countershock protection. The monitor-recorder shall be protected from the defibrillator energy output.

3.12.12 ECG Calibration. ECG calibration shall be accomplished by a 1 mV reference signal.

3.12.13 Stylus. The recorder shall have an operator adjustable and changeable recording stylus or thermal array recording device.

3.12.14 Paper speed. The recorder shall have a paper speed of 25 mm/second. Paper speed shall be accurate to within plus or minus 5 percent.

3.12.15 Printer output. The printer shall produce a hard copy report on ECG paper.

3.12.16 Modularity. The monitor-recorder shall be an integral part of the defibrillator system but may be detachable, in which case it shall be capable of operating on a separate internal battery.

3.12.17 Common mode rejection ratio. The common mode rejection ratio (CMRR) shall be 80 db minimum with respect to chassis ground with 5K ohms misbalance at 60 Hz.

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3.12.18 Freeze control. The monitor-recorder shall have a freeze control or a delayed recording capability for detailed analysis of ECG arrhythmias.

3.12.19 Paper size. ECG paper shall have a minimum grid width of 40mm and contain at least 50 feet per roll. A low paper indication shall be provided, either by the instrument or in the form of a mark on the paper.

3.12.20 Recorder integrity. The recorder shall be an integral part of the monitor of the defibrillator/monitor-recorder system and not a stand-alone instrument requiring electrical cabling interconnection to provide patient data.

3.12.21 Patient cable leads. The patient cable leads supplied with the monitor shall be color coded in conformance with AHA publication, "Recommendation for Standardization of Leads and of Specification for Instruments in Electrocardiography and Vectorcardiography".

3.12.22 Performance standards. Cardioscope, recorder, and ECG amplifier performance requirements shall be in accordance with "Standard for Cardiac Monitors, Heart Rate Meters and Alarms, " ANSI/AAMI EC13-1983 (unless otherwise specified).

3.12.23 Cardioscope-recorder interface. The cardioscope-display shall interface with the recorder to allow the recorder to document real time while the monitor is in the freeze mode.

3.12.24 Syncronization marker. The monitor shall generate a syncronization marker that will appear on the cardioscope when syncronized cardioversion is attempted.

3.13 Electrical characteristics. The ECG monitor-recorder and defibrillator shall operate from self-contained batteries. The system shall be capable of providing all functions specified herein. Defibrillator and ECG monitor-recorder modules that can be used independently shall be independantly powered, each containing an internal battery power source.

3.13.1 DEPMEDS requirements. The electrical characteristics of the battery charger shall be consistent with the requirements contained in the latest edition of MIL-D-42048.

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3.13.2 Battery capacity. The defibrillator/monitor-recorder shall be capable of full operation while powered by internal batteries for 2.25 continuous hours of cardioscope monitoring, 45 continuous minutes of recorder operation (or any linear combination), 30 defibrillator counter shocks at maximum energy into a 50 ohm load at an ambient temperature of 82.4°F (28°C), and 10 defibrillator counter shocks at maximum energy into a 50 ohm resistive load within 5 minutes at an ambient temperature of 32°F (0°C).

3.13.3 Charge time to Emax. Charge time to maximum energy output shall not exceed 12 seconds at 82.4°F (28°C).

3.13.4 Charge time to full charge. The charge time for fully depleted batteries to fully charged state shall not exceed sixteen hours. The unit shall have a level indicator to show the level to which the batteries have been charged after the sixteen hours.

3.13.5 Charge indication. A charging lamp/indicator shall be provided for each charge circuit which is illuminated during the periods when the battery charger is functioning.

3.13.6 Low voltage indication. A low voltage indicator shall be provided on the front of the defibrillator and monitor-recorder for each battery unit.

3.13.7 Battery packs. The defibrillator/monitor-recorder shall allow rapid interchange (operator instalable/cassette type) of battery packs, and be provided with two spare battery packs in addition to the batteries supplied with the unit.

3.14 Accessories. The defibrillator/monitor-recorder system shall be provided with but not limited to the accessories described herein.

3.14.1 Patient cables. Two patient cables shall be supplied.

3.14.2 Spare stylus. One spare stylus shall be supplied (with replacement tool, if required).

3.14.3 ECG ruler. One ECG ruler shall be supplied.

3.14.4 Pregelled electrodes. A minimum of twelve pregelled electrodes and four metal patient electrodes with electrode straps, and one suction electrode shall be supplied.

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3.14.5 Spare batteries. Two spare batteries of each type shall be supplied.

3.14.6 Recording paper. Three packages of recording paper shall be supplied.

3.14.7 Electrode jelly. One tube of electrode jelly shall be supplied.

3.14.8 Accessory bag. One accessory bag shall be supplied.

3.14.9 External adult paddles. One pair of external adult paddles shall be supplied.

3.15 Additional item. When specified by the contracting officer external pediatric paddles, providing at least a 15 square centimeter contact area per electrode, shall be provided.

3.16 Drip proof test. The defibrillator/monitor-recorder system shall be capable of withstanding the drip proof test specified herein without damage and shall operate properly in all modes following the test. All controls must be immediately accessible (no cover) during all steps of the test.

3.17 Workmanship. The defibrillator/monitor-recorder shall be free from defects which detract from its appearance or impair its serviceability.

3.18 Unit. Each (EA). One defibrillator/monitor-recorder, as specified, constitutes one unit.

3.19 Service data. The contractor shall enclose two copies each of operating instructions, maintenance instructions, price parts list, and wiring schematic/diagrams.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.3 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.4 First article inspection. First article inspection shall be performed on the defibrillator system when a first article sample is required. This inspection shall include examination and tests.

4.5 Sampling.

4.5.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one defibrillator system.

TABLE I. Sampling for examination.

	Inspection level	AQL (defects per hundred units)
Major defects	II	1.0
Total defects (Major and minor combined)	II	4.0
For dimensional examination	S-1	2.5
For All Tests	S-2	1.5

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4.5.2 For tests. Sampling for tests shall be conducted as follows:

4.5.2.1 First article. Where a first article sample is required, all tests shall be performed on that unit.

4.5.2.2 Acceptance testing. When a first article sample has not been required, one unit shall be subjected to the performance test specified herein. If a first article sample has successfully passed all tests, performance test shall not be required at acceptance.

4.6 Examination. The defibrillator/monitor-recorder shall be examined to determine compliance with the requirements of this specification in accordance with the classification of defects in table II.

TABLE II. Classification of Defects *

Classification	Defects
Major	
101	Component missing.
102	Component damaged including (but not restricted to) cracks, checking, breaks and deformed areas.
103	Exposed sharp edges or corners accessible to operator during normal use or cleaning.
104	Synchronization switch not recessed or protected from accidental engagement.
105	Power switch and charge switch/button not clearly labeled and readily visible.
106	Paddle storage not neat and "out of the way".
107	Modules do not fit together correctly.
108	Any knobs, switches, or buttons missing or misaligned.
109	Stylus bent or misaligned.
110	Operator and service manuals not supplied.
Minor	
201	Identification marking not complete, permanent, and legible.
202	Finish not free of chips, cracks, scratches, and other appearance defects.
203	Any component part not secured in place.

*Examination shall not be limited to the classified possible defects listed herein.

4.7 Tests. Performance and acceptance tests shall be conducted.

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4.7.1 Performance. The defibrillator, monitor-recorder shall be completely assembled in its operating configuration and tested to determine compliance with this specification. Testing shall include Acceptance Testing.

4.7.2 Acceptance testing. The defibrillator/ECG monitor-recorder must pass the tests as described in table III.

Table III. Tests.

I.	Strain Relief (Battery Charger Line Cord) Gradually apply a 35 lb. pull in a direction that would be the most likely to pull the cord from the enclosure and maintain for one minute. The strain relief is not acceptable if there is such movement of the cord as to indicate that stress is being transmitted to terminals, splices or interior wiring.
II.	Air Clearance Spacing Measurements shall be made, using an appropriate measuring instrument, to verify isolated patient connected circuitry is spaced at a minimum of 1/8 inch from non-isolated circuitry.
III.	Leakage current: A. Defibrillator 1. Chassis Leakage: Leakage current measured between any exposed conductive surface and ground shall not exceed 100 microamperes. 2. Paddle Leakage: Leakage current measured between an external paddle electrode and ground shall not exceed 10 microamperes. Leakage current measured between an internal paddle electrode and ground shall not exceed 10 microamperes. 3. Conditions: Leakage current measurements will be made under all of the following conditions. a) Power on and off. b) Defibrillator in both charged and uncharged state.

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Table III. Tests - Continued.

B. ECG Monitor/Recorder

1. Chassis Leakage:
Leakage current measured between any exposed conductive surface and ground shall not exceed 100 microamperes

2. Lead leakage (source current)leakage current · measured between each lead and ground, all leads ground, and between any two leads shall not exceed 10 microamperes.

3. Lead leakage (sink current):
Sink current measured between each load at a potential of 120V and ground shall not exceed 20 microamperes. This test shall be performed under the conditions listed above except the ground remains intact.

C. Test Device

All leakage current measurement will be performed with a test device equipped per ANSI/AAMI ES1-1985, American National Standard, Safe Current Limits for Electromedical Apparatus.

IV. Defibrillator Paddle Insulation

A. Apply aluminum foil to the entire gripping surface of one paddle with sufficient pressure to make it conform to the surface.

NOTE: The gripping surface for verification purposes is defined as the approximately cylindrical area around each handle where the operator's fingers grasp the paddle, but not beyond any barrier or curvature that is obviously intended to limit hand placement, including areas one inch in diameter centered on any control located on the paddle, representing the location(s) of the operator's thumb and/or finger.

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Table III. Tests - Continued.

B. With the paddles connected to the defibrillator, connect the high potential lead of a 60 Hz (or 50 Hz, if preferred) high potential test source to the aluminum foil. Connect low-potential or ground lead to the paddle electrode. Slowly adjust the peak voltage amplitude of the test source to equal 1.5 times the maximum voltage occurring on the defibrillator energy storage capacitor. For a sine wave this corresponds to an rms value of 1.06 times the capacitor voltage. Maintain this voltage for 1 minute. Repeat this test with the low-potential lead connected to the other paddle electrode.

C. There shall be no voltage breakdown, and the 60 Hz leakage current shall not exceed 250 microamperes peak (177 microamperes rms); if a 50 Hz test source is used, the current shall not exceed 210 microamperes peak (148 microamperes rms).

NOTE: Any leakage current that flows when the high-potential lead is disconnected from foil, but the output voltage is set as described in (B) represents internal leakage of the test apparatus and may be subtracted from the total current reading. Alternatively, an external ammeter may be placed in the ground circuit to eliminate this internal leakage and to enhance resolution, if necessary.

D. Repeat (A) through (C) using the other paddle grip.

V. ECG Monitor-Recorder Sweep/Chart Speed

Measure the monitor sweep speed and the recorder paper speed by applying a signal of known frequency to the patient cable. Monitor sweep speed shall be within plus or minus 10 percent of 25 mm/sec. and recorder chart speed within plus or minus 5 percent of 25 mm/sec.

VI. ECG Amplifier Calibration

The accuracy of the 1 mV internal calibration pulse shall be measured by comparing the amplitude of this pulse to that of a calibrated 1 mV pulse applied to the patient cable from a pulse generator. It shall be accurate to plus or minus 15 percent.

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Table III. Tests - Continued.

VII. ECG Amplifier Common Mode Rejection

Common mode rejection is measured by comparing the amplitude of a 1 mV differential signal applied to the monitor to the amplitude of a higher voltage sinusoidal signal of 60 Hz applied between the junction of the leads and ground. Measured common mode rejection shall be at least 80 dB with respect to chassis ground with 5K ohms imbalance between leads.

VIII. Defibrillator Overload Protection

Overload protection shall comply with those requirements specified in ANSI/AAMI EC 13.

IX. ECG Monitor and Recorder Frequency Response

Apply a 1 mV peak-to-peak differential sinusoidal signal to the patient cable, and vary the frequency. The bandwidth is defined by the upper and lower frequencies at which the trace deflection amplitude decreases to 0.707 times the value. This bandwidth maximum shall be 0.5 Hz to anywhere between 35 to 50 Hz inclusive, on the monitor and recorder, real time mode.

X. Defibrillator Energy Accuracy

Connect the defibrillator paddles to a defibrillator analyzer equipped with a 50 ohm resistive load. Charge the defibrillator to a preselected energy level, then discharge into the analyzer. Repeat for each available energy level setting.

XI. Defibrillator Charge Time

Using a stopwatch, measure the time between activation of the charge control and full charge indication for the first and last of 10 sequential charges and discharges all at maximum energy. This time shall not exceed 12 seconds.

XII. Defibrillator Discharge Pulse Rate

Apply the paddle electrodes to a defibrillator tester and select the maximum energy level. Simultaneously initiate the charge and a timer.

Immediately upon completion of the charge, discharge the defibrillator.

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Table III. Tests - Continued.

The defibrillator shall be capable of delivering 10 pulses at maximum energy, into a 50 ohm resistive load within 5 minutes, at 0°C.

XIII. Defibrillator Overload Protection

Overload protection shall comply with those requirements specified in ANSI/AAMI EC 13, except the defibrillator energy level shall be 400J.

XIV. Synchronized Discharge

Connect a signal generator to the ECG monitor. Measure the time between the leading edge of square pulse and the leading edge of the defibrillator's synchronized discharge pulse. This time can be displayed automatically on a defibrillator analyzer if so equipped, or measured directly from an oscilloscope, and shall be within 30 milli-seconds.

XV. Battery Capacity

Battery Capacity will be tested by verifying 2.25 hours of continuous cardioscope monitoring or 45 minutes of continuous recording of a heart rate of 60 BPM. Defibrillator battery capacity will be tested by verifying 30 discharges at maximum energy into a 50 ohm load at an ambient temperature of 82.5°F (28°C).

XVI. Environmental Safety/Drip-Proof Test

The contractor shall be responsible for the performance of drip-proof test as follows:

Step 1. Place item in its normal operating position, with patient cable and defibrillator paddles attached, on a platform or tilt table in a manner such that it can be oriented horizontally and in any of four directions forward, backward, and to each side to the extent of 15 degrees from the horizontal plane.

Step 2. Attach a spray nozzle with a uniformly distributed full cone spray pattern to a flexible hose and using a container calibrated in liters or an equivalent method, determine the rate of flow out of the nozzle. The requirements for water flow shall be 16.3 plus or minus 1 liter (4.3 gallons) per hour for each square foot of area covered by the spray 0.9 m (3 feet) below the nozzle.

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Table III. Tests - Continued.

Step 3. Position the nozzle 0.9 m (3 feet) above the item so that the spray is directed downward onto it. The item shall be subjected to a uniform spray over the entire surface being tested. During the test, the item shall not be operated, but all controls shall be immediately accessible to the operator: i.e., uncovered, with the exception of the recorder, which may be protected.

Step 4. Direct the spray onto the top of the item (with platform or tilt table horizontal) for at least eight minutes. Turn off spray four minutes.

Step 5. Direct the spray downward onto the item for at least eight minutes with the platform or table tilted such that the item is oriented 15 degrees from the horizontal in the forward position. Turn off spray four minutes.

Step 6. Direct the spray downward onto the item for at least eight minutes with the platform or table tilted such that the item is oriented 15 degrees from the horizontal in the backward position. Turn off spray four minutes.

Step 7. Direct the spray downward onto each remaining side of the item for at least eight minutes with the platform or table tilted such that the side being sprayed is oriented 15 degrees from the vertical. Turn off spray for four minutes.

Step 8. Return the item to the horizontal and perform a satisfactory operation test.

XVII. Humidity

Manufacturer shall supply the following, verifying compliance with the test described below:

a. Copy of test procedure as specified below in test method.

b. Unit type.

c. Serial number.

d. Manufacturer and model number of chamber used.

e. Leakage current test results after 48 hours.

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Table III. Tests - Continued.

f. Energy accuracy test results after 48 hours. g. Test Method 1. Place the unit in test chamber such that leakage current and delivered energy accuracy tests can be performed external to the chamber. 2. Adjust the temperature of the chamber to 82°F(28°C). 3. Permit sufficient time for the temperature "inside the unit" to stablize prior to proceeding. 4. After temperature stabilization, adjust the R.H. of the chamber to 95 (plus or minus 5) percent and permit sufficient time for humidity stabilization. 5. After 48 hours of conditioning, source and sink leakage current tests and energy accuracy tests are to be performed. The unit is in compliance if leakage currents and energy accuracy are within the specified limits.

5. PACKAGING

5.1 General.

The complete requirements for packaging shall consist of this specifications and the latest issue of Military Specification MIL-D-42048 except where otherwise indicated.

6. NOTES

6.1 Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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6.2 Regulatory requirements.

6.2.1 Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

6.2.2 Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

6.2.3 Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order. Provisioning technical documentation required and appropriate Data Item Descriptions (DID's) should be included as part of the contract.

6.2.4 Subject term key word listing.

Defibrillator
Monitor
Recorder
Cardioscope
Battery operated
With charger

6.2.5 Applicable NSN. This specification covers the requirements for the following item:

NSN	Item Identification
6515-01-204-1854	Defibrillator/Monitor-recorder battery operated, with charger

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Custodians:

Air Force - 03

Review Activities:
Army - MD
Navy - MS

User Activities:
Air Force - 03

Preparing activity:

DOD-MB

Agent:

DLA-DM

Project No. 6515-4414

Location: MILSP/Defib/
Mon/Rec/H17-15

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-42054A	DEFIBRILLATOR, MONITOR-RECORDER (Battery Operated, with Charger)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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